Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2023-P2-Model)** **Date: 10-05-2026**

Time: 02.30Pm to 05.30Pm **GTA-31** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**A**	2	**D**	3	**D**	4	**A**	5	**ABD**	6	**CD**
7	**ABD**	8	**0**	9	**1**	10	**11**	11	**179**	12	**8**
13	**5**	14	**3**	15	**1**	16	**5**	17	**4**		

PHYSICS

18	**B**	19	**B**	20	**A**	21	**C**	22	**BCD**	23	**ACD**
24	**ACD**	25	**3**	26	**9**	27	**5**	28	**4**	29	**1**
30	**5**	31	**3.41**	32	**5**	33	**3.46**	34	**2**		

CHEMISTRY

35	**C**	36	**B**	37	**C**	38	**B**	39	**BCD**	40	**BD**
41	**ABC**	42	**6**	43	**72**	44	**268**	45	**24**	46	**84**
47	**1200**	48	**6**	49	**6**	50	**2.5**	51	**12.5**		

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SOLUTIONS
MATHEMATICS

1. $\vec{c} = \lambda\vec{a} + \mu\vec{b}$

Dot with $\vec{a},\vec{b}$ on both sides

Solve two linear equations to get $\lambda \& \mu$.

2. As the magnitudes of slopes are equal draw TH parallel axis

$RT = RH$

TH parallel axis $\Rightarrow$ H is the mid point of PQ

WKT, ST is angular bisector of $\angle QSP$ $\frac{PR}{RQ} = \frac{a}{b} = \frac{a\lambda}{b\lambda}(say)$

$PR = a\lambda,\ RQ = b\lambda,\ PQ = (a+b)\lambda$ $\Rightarrow\ RH = a\lambda - \frac{a+b}{2}\lambda = \frac{a-b}{2}\lambda = RT$

And perimeter of $\Delta PQS = a+b+(a+b)\lambda\ = (a+b)(\lambda+1)$

3. $p = \frac{2^{2\times 2}}{2^{3\times 3}} = \frac{1}{32}$ $\quad p = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Total number of matrices = 2^9

$(s_1s_2s_3t_1t_2t_3)$ is odd integer if

1) all the elements are 1 (no of matrices = 1)

2) four zeros and five ones (no of matrices = 9)

(zeros must occupy all the positions of any minor)

3) six zeroes and three ones (no of matrices = 6)

$\left({}^3C_1.{}^2C_1.1\right)$ $\Rightarrow$ required probability $= \frac{16}{2^9} = \frac{2^y}{2^9}$

4. The given equation can be written as,

$$4^x \cdot 4^{3/2} + 9^x \cdot 9^{1/2} = 10 \cdot 6^x \Rightarrow 8 \cdot 4^x + 3 \cdot 9^x = 10 \cdot 6^x$$

$$\Rightarrow 8 \cdot \left(\frac{2}{3}\right)^x + 3 \cdot \left(\frac{3}{2}\right)^x = 10$$

[Dividing throughout by 6^x]

$\Rightarrow\ 8y + \frac{3}{y} = 10$

$\Rightarrow\ 8y^2 - 10y + 3 = 0$

$\Rightarrow\ y = \frac{10 \pm \sqrt{4}}{16} = \frac{3}{4}, \frac{1}{2} \Rightarrow \left(\frac{2}{3}\right)^x = \frac{3}{4}, \frac{1}{2} \Rightarrow \left(\frac{3}{2}\right)^x = \frac{4}{3}, 2$

$\Rightarrow\ x\log\left(\frac{3}{2}\right) = \log\left(\frac{4}{3}\right), \log 2$ $\Rightarrow\ x = \frac{\log\left(\frac{4}{3}\right)}{\log\left(\frac{3}{2}\right)}, \frac{\log 2}{\log\left(\frac{3}{2}\right)} = \log_{1.5}\left(\frac{4}{3}\right), \log_{1.5} 2$

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5. $$\begin{bmatrix} x^2 \\ y^2 \\ z^2 \end{bmatrix} = \begin{bmatrix} \frac{2b^2+2c^2-a^2}{4} \\ \frac{2c^2+2a^2-b^2}{4} \\ \frac{2a^2+2b^2-c^2}{4} \end{bmatrix} = \frac{1}{4}\begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}\begin{bmatrix} a^2 \\ b^2 \\ c^2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} a^2 \\ b^2 \\ c^2 \end{bmatrix} = 4\begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}^{-1}\begin{bmatrix} x^2 \\ y^2 \\ z^2 \end{bmatrix} = \frac{4}{9}\begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}\begin{bmatrix} x^2 \\ y^2 \\ z^2 \end{bmatrix}$$

6. $f''(x) > 0 \ \forall x \in [0,2] \qquad \Rightarrow f'(x) \uparrow$

$f'(c_1) = \frac{f(1)-f(0)}{1-0}, c_1 \in (0,1)$ and $f'(c_2) = \frac{f(2)-f(1)}{2-1}, c_2 \in (1,2)$

$f'(c_1) < f'(c_2) \Rightarrow f(0)+f(2) > 2f(1)$

Similarly applying LMVT between $\left[0, \frac{2}{3}\right]$ and $\left[\frac{2}{3}, 2\right]$

$$\frac{f(2)-f\left(\frac{2}{3}\right)}{\frac{4}{3}} > \frac{f\left(\frac{2}{3}\right)-f(0)}{\frac{2}{3}} \Rightarrow 2f(0)+f(2) > 3f\left(\frac{2}{3}\right)$$

7. Tangents drawn form $(-2,0)$ to $y^2 = 4x$ will have slopes $\pm\frac{1}{\sqrt{2}}$.

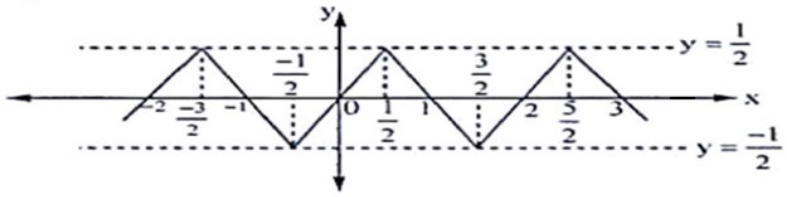
And tangents drawn from $(-2,0)$ to $x^2+y^2-4x-4y=0$ will have slopes $1 \pm \sqrt{\frac{3}{2}}$.

For $a = 1-\sqrt{\frac{3}{2}}, \frac{1}{\sqrt{2}}, 0, \frac{2}{3},$ no. of points of intersection is 4.

8. $$f(x) = \begin{cases} x, -\frac{1}{2} \le x < \frac{1}{2} \\ 1-x, \frac{1}{2} \le x < \frac{3}{2} \\ x-2, \frac{3}{2} \le x < \frac{5}{2} \\ 3-x, \frac{5}{2} \le x < \frac{7}{2} \end{cases}$$

And so on. Graph of $f(x)$ will be as shown below

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Clearly $f(x)$ is an odd function.

$\cos(\pi x)$ is an even function, so $f(x).\cos(\pi x)$ will be an odd function

Hence $\frac{\pi^2}{5}\int_{-20}^{20} f(x)\cos(\pi x)=0$

9. $z=x+iy$ $\quad |6x+8y+4|+|8x+6y+6|=10$

$\Rightarrow 7x+8y=0, 7x+8y=-10, x-y=4, x-y=-6$

Has $(-1,0)$ inside it & $(0,1)$ outside it.

$|z+1|+|z-i|=\sqrt{2} \Rightarrow y=x+1; x\in[-1,0]$

10. $P(x)-5=(x-2)Q(x)$ $\quad P(x)-3=(x-4)R(x)$

Let P(x) has one integral root n such than P(n)=0

P(n) – 5 = (n - 2) Q(n) ……(i)

$-\frac{5}{n-2}=Q(n)$

As Q(n) is also an integer

So n – 2 can be $\pm 1 and \pm 5$ only

$n-2\Rightarrow n=3$ $\quad n-2=1\Rightarrow n=1$

$n-2=5\Rightarrow n=7$ $\quad n-2=-5\Rightarrow n=-3$

Are possible values of n

$P(n)-3=(n-4)r(n)$

$-\frac{3}{n-4}=r(n)$

Hence n – 4 can be $\pm 1 and \pm 3$

$n-4=1\Rightarrow n=5$ $\quad n-4=-1\Rightarrow n=3$

$n-4=3\Rightarrow n=7$ $\quad n-4=-3\Rightarrow n=1$

So possible value of n should satisfy both equation hence

Possible values of n are 1, 3, 7, So, (c) is incorrect option

11. $(n-1)a_n=a_1+a_2+.....+a_{n-1}\ \forall n\geq 3$

$n\,a_{n+1}=a_1+a_2+.....+a_n$

$n\,a_{n+1}-(n-1)a_n=a_n$ $\quad a_{n+1}=a_n$ $\quad \forall n\geq 3$

12. $\sum_{r=1}^{6} {}^7c_r a^{7-r}b^r=\left(a^7+b^7\right)-a^7-b^7$ ……..1

$\frac{(a+b)^7-a^7-b^7}{ab(a+b)\left(a^2+ab+b^2\right)^2}=7$ ……2

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$$7ab(a+b)\left(a^2+ab+b^2\right)^2=7^3$$

Divide1/2, we get

$\left(a^2+ab+b^2\right)$ is divisible by 7^2

$a^2+ab+b^2=49\Rightarrow a=5,b=3$

13. Put $z=-1$ in (A) : $1=|1+a_1||1+a_2|.....|1+a_{24}|$

$$1=2^{24}\left(\cos\frac{\pi}{25}\cos\frac{2\pi}{25}.....\cos\frac{24\pi}{25}\right)$$

$$1=2^{24}\left(\cos\frac{\pi}{25}\cos\frac{2\pi}{25}.....\cos\frac{12\pi}{25}\right)^2$$

Divided (i) and (i) ans 5

$$z^{25}-1=(z-1)(z-\alpha)\left(z-\alpha^2\right)....\left(z-\alpha^{24}\right)$$

$$\frac{z^{25}-1}{z-1}=(z-\alpha)\left(z-\alpha^2\right)....\left(z-\alpha^{24}\right)$$

Put 2 = 1

$$25=|(1-\alpha_1)||(1-\alpha_2)||(1-\alpha_3)|.....|(1-\alpha_{24})|$$

$$25=2^{24}\sin\frac{\pi}{25}\sin\frac{2\pi}{25}\sin\frac{3\pi}{25}.....\sin\frac{24\pi}{25}$$

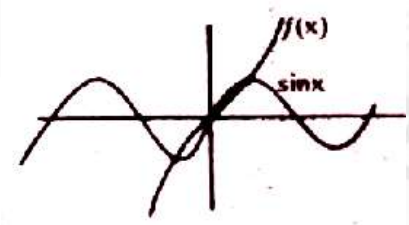
$$25=2^{24}\left(\sin\frac{\pi}{25}\sin\frac{2\pi}{25}.....\sin\frac{12\pi}{25}\right)^2$$

14, 15.

$$\frac{dy}{dx}=|y|+1\Rightarrow\frac{dy}{y+1}=dx,y\geq0\quad or\frac{dy}{-y+1}=dx,y<0\qquad\therefore y\begin{cases}e^x-1, & x\geq0\\ 1-e^{-x}, & x<0\end{cases}$$

We get $f(x)=-f(-x)\Rightarrow f(x)$ is odd function

There are 3 solutions $\qquad a_1\in(-1,0)\qquad a_2=0\qquad a_3\in(0,1)$



16, 17.

$(2019)^2=3^2\times(673)^2$ is L.C.M of a, b, c, d

In $\left(3^4-2^4\right)^2=4225=5^2\times13^2$ *ways*

n=4225 is L.C.M of (a_1,b_1,c_1,d_1) in 4225 ways

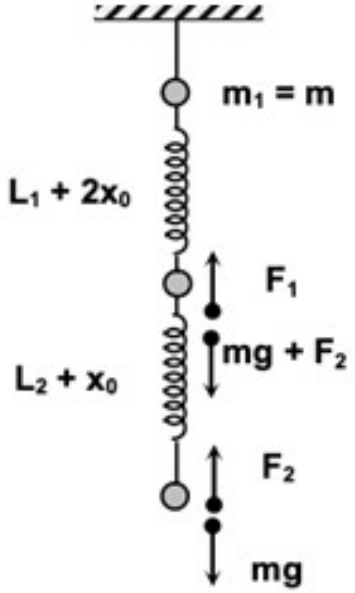
$n_1=n_2=n_3.....n_{10}=n=4225$

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PHYSICS

18. Since tensions in the 2 springs are different so they have different natural lengths. Let the natural length of upper and lower spring be L_1 and L_2 respectively.

Since, in equilibrium



$\therefore\ F_2 = kx_2 = mg$

$\Rightarrow\ x_2 = \dfrac{mg}{k} = x_0\,(say)$

And $F_1 = mg + F_2 \Rightarrow F_1 = 2mg$

$\Rightarrow\ kx_1 = 2mg \Rightarrow x_1 = \dfrac{2mg}{k} = 2x_0$

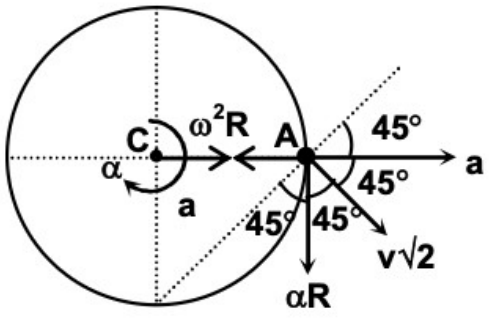
Given $L_1 + 2x_0 = L_2 + x_0 \Rightarrow L_2 = L_1 + x_0$

When the springs are at their natural lengths, the distance of centre of mass from top ball will be

$$X_{CM} = \frac{0 + mL_1 + m(L_1 + L_2)}{3m} = L_1 + \frac{x_0}{3}$$

$\therefore$ CM is at a distance $\dfrac{x_0}{3} = \dfrac{mg}{3k}$ below the second ball.

19. Velocity of point $'A'V_A = \sqrt{V^2 + \omega^2 R^2} = V\sqrt{2}$



Normal acceleration of point A,

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$$a_A(n) = \omega^2 R\cos 45° + \alpha R\cos 45° - a\cos 45°$$

$$a_A(n) = \frac{\omega^2 R}{\sqrt{2}} = \frac{V^2}{\sqrt{2}R}$$

$\therefore$ radius of curvature of trajectory of point 'A' relative to the ground is

$$r = \frac{(V_A)^2}{a_{A(n)}} = \frac{\left(V\sqrt{2}\right)^2}{\frac{V^2}{\sqrt{2}R}} = 2\sqrt{2}R$$

20. By conservation of energy $\frac{-GMm}{R} + \frac{1}{2}mv'^2 = \frac{-GMm}{R} + \frac{1}{2}mv_0^2 \ldots\ldots(i)$

By conservation of angular momentum $mv_0 2R = mv'R$

$\Rightarrow v' = 2v_0 \ldots\ldots(ii)$

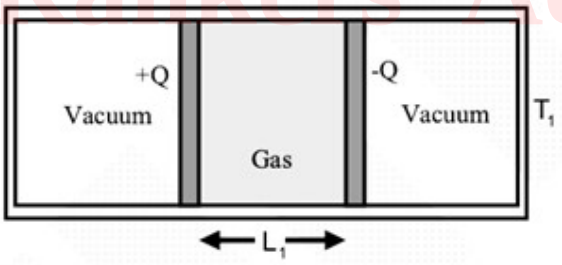
$\Rightarrow v_0 = \sqrt{\frac{GM}{3R}}$ (using (i) & (ii))

For circular orbit $\frac{mv''^2}{2R} = \frac{GMm}{(2R)^2} \Rightarrow v'' = \sqrt{\frac{GM}{2R}}$

$$v'' = \sqrt{\frac{3}{2}}v_0$$

21. Centre of application of electric force will be the centre of mass.

22. Stage 1

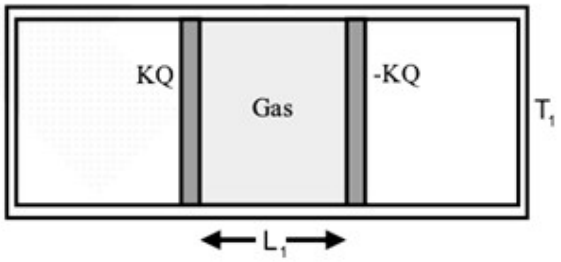


$P_1 AL_1 = nRT_1$

$\frac{Q^2}{2A\varepsilon_0} = P_1 A = \frac{nRT_1}{L_1}$ (Force Balancing)

Stage 2

Charge is suddenly changed



System is not in equilibrium

Stage 3

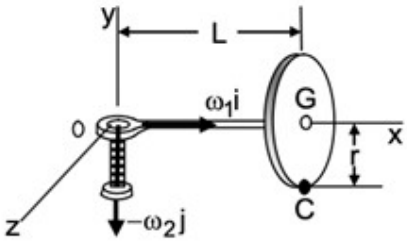
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23. $1000 = \dfrac{320}{4l_1} \Rightarrow l_1 = 8\,cm$

$1000 = \dfrac{3\times 320}{4l_2} \Rightarrow l_2 = 24\,cm$

$-A\dfrac{dh}{dv} = 2\sqrt{2gh} \Rightarrow -\int \dfrac{dh}{\sqrt{h}} = \dfrac{a}{A}\sqrt{2g}\int dt$

24. $\omega = \omega_1 i - \omega_2 j$



$V_C = \omega \times r_C = 0$

$(\omega_1 i - \omega_2 j)\times(Li - rj) = 0$

$(L\omega_1 - r\omega_2)k = 0 \qquad \omega_2 = \omega_1 i / L$

$\omega = \omega_1 i - (r\omega_1 / L)j$

$H_x = I_x\omega_x = \left(\dfrac{1}{2}mr^2\right)\omega_1$

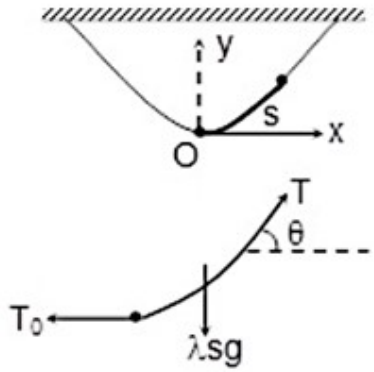
$H_z = I_z\omega_z = \left(mL^2 + \dfrac{1}{4}mr^2\right)0 = 0$

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25. $T = \sin\theta = \lambda sg \quad .....(1)$

$T = \cos\theta = T_0 \quad .....(2)$

(1) and (2) $\tan\theta = \lambda sg / T_0 \Rightarrow S = \left(\dfrac{T_0}{\lambda g}\right)\tan\theta.$



$R.O.C = \dfrac{ds}{d\theta}$

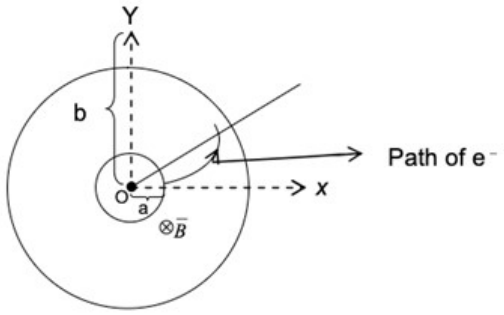
$\Rightarrow R = \left(\dfrac{T_0}{\lambda g}\right)\sec^2\theta$

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Putting $T_0 = 90; \lambda = 4; g = 10 \, and \, \theta = 30°$ $R = 3m$

26. $m\left\{r\frac{d^2\theta}{dt^2} + 2\left(\frac{dr}{dt}\right)\left(\frac{d\theta}{dt}\right)\right\} = eB\left(\frac{dr}{dt}\right)$

$\Rightarrow \frac{d}{dt}\left(r^2\theta\right) = \frac{eB}{m} r\, r$ $\frac{d}{dt}\left(r^2\theta\right) = \frac{eB}{2m} r\left(\frac{d}{dt} r^2\right)$



$\Rightarrow r^2\theta = \left(\frac{eB}{2m}\right)\left(r^2 - a^2\right) \quad \left\{\frac{d\theta}{dt} = 0 \, for \, r = a\right\}$

$\& \ \theta = \frac{eB}{2m}\left(1 - a^2/b^2\right)$

$v^2 = \frac{2eV}{m} = \left(\frac{e^2B^2}{4m^2}\right) b^2\left(1 - a^2/b^2\right)$

Put the value & get the result.

27. $\Delta E = 3.4eV - \left(-13.6\frac{z^2}{n^2}\right)$ $= 17\,eV$

$\frac{hc}{\lambda_1} = 17\,eV _____ (1)$

2nd condition = $\Delta E' = E_2 - E_1$

$= \frac{-13.6 \times 4}{22}\left(+\frac{13.6 \times 4}{1}\right)$

$\Delta E' = 40.8\,eV$

$\Delta E' = \frac{hc}{\lambda_2} = 40.8\,eV$

$\frac{\lambda_1}{\lambda_2} = 2.4$

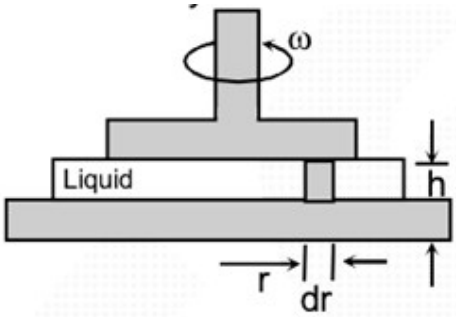
28. Consider an element of disc at a radius r and having a width dr.
Linearly velocity at this radius = ωr

Shear stress, $\tau = \mu\frac{du}{dy}$

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Assuming, the gap h to be small, so that the velocity distribution may be assumed linear

$$\tau = \mu \times \frac{v}{h} = \mu \frac{\omega r}{h}$$



Viscous force, $dF = \tau \times Area$

$$= \tau \times 2\pi r dr$$

Torque dT on the element,

$$dT = dF \times r = \tau 2\pi r^2 dr$$

Or $dT = \frac{\mu\omega r}{h} \times 2\pi r^2 dr = \frac{2\pi\mu\omega r^2 dr}{h}$

Total torque, $T = \int_0^{d/2} \frac{2\pi\mu\omega r^3 dr}{h} = \frac{\mu\pi d^4 \omega}{4h}$

Thus, $X = 4$.

29. $w = \frac{\lambda D}{d} = 1\text{mm}$

$$\frac{y}{D} = \frac{y_1}{2D}$$

$$y_1 = 2y = \frac{1}{2}\text{mm}$$

$$y = \frac{1}{2}\sin\frac{\pi}{6}t$$

$$\Rightarrow t = 1\text{sec}$$

30. $v_0 = \omega h \sin\theta$

$$\Rightarrow \omega = \frac{v_0}{h\sin\theta} = \frac{2v_0}{h} = 5\text{rad/s}$$

31. Consider the case where there is current coming in just at A. Then, J always points radially outward from A and has (hemi) spherical symmetry, with magnitude

$$J.2\pi r^2 = I_0 \Rightarrow J = \frac{I_0}{2\pi r^2}.$$

Then we have $E = \frac{I_0}{2\pi\sigma r^2}\hat{r}$

Where σ is the conductivity of the material, which implies

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$$V_D - V_C = \int_a^{\sqrt{2}a} \frac{I_0}{2\pi\sigma r^2} dr = \frac{I_0}{2\pi\sigma a}\left(1 - 1/\sqrt{2}\right).$$

Similarly, for the case where the current is coming out of B, we have

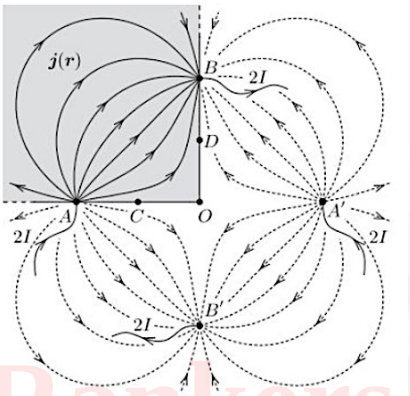
$$V_D - V_C = \frac{I_0}{2\pi\sigma a}\left(1 - 1/\sqrt{2}\right).$$

The actual voltage drop is the superposition of the two, $\Delta V = \frac{I_0\rho}{2\pi a}\left(2 - \sqrt{2}\right)$

Where $\rho = 1/\sigma$ is the resistivity. Then ρ can be calculated as

$$\rho = \frac{2\pi a \Delta V}{I_0\left(2 - \sqrt{2}\right)} = \frac{\pi a\left(2 + \sqrt{2}\right)\Delta V}{I_0}$$

32.



We have a new current source and sink respectively at the reflections of A and B in O. the current sources and sinks all have magnitude 2I, rather then I, because only half of the currents at A and B actually enter and exit the physical plate, shaded in gray,

Now, if a current 2I enters the plate, the current at a distance r is $\frac{2I}{2\pi rt}$, so the electric field at a distance r is $\frac{\rho I}{\pi rt}$ (pointing radially outward), so the potential is $\frac{\rho I}{\pi t}\log\left(r_0 / r\right)$ for some arbitary r_0, which well take to be the same for all current sources and sinks. Then

$$V_C = \frac{\rho I}{\pi t}\left(\log\left(r_0/d\right) + \log\left(r_0/3d\right) - 2\log\left(r_0/\sqrt{5}d\right)\right) = \frac{\rho I}{\pi t}\log\left(5/3\right).$$

By symmetry, $V_C - V_D$. Thus,

$$V = 2V_C = \frac{2\rho I}{\pi t}\log\left(5/3\right), \qquad \rho = \frac{\pi t}{\log\left(5/3\right)}\frac{V}{I}.$$

33. For charge density to be zero, electric field must be parallel to the plate.

$$\theta = \tan^{-1}\left(\sqrt{2}\right) \qquad r = \sqrt{3}h = 3.46m$$

34. $$U = \frac{1}{2} \times \frac{2Kp^2}{(2h)^3}$$

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CHEMISTRY

35. $\Rightarrow$ Basic character of hydroxides of tripositive lanthanoid cation decreases from Ce to Lu .

36. With increasing the stability of carbocation intermediate, percentage of racemisation increases. With increasing the concentration of the nucleophile (H_2O), percentage of inversion increases and hence percentage of racemisation decreases
a) 34% racemisation
b) 96 % racemisation
c) 80 % racemisation

37. A) Aryl halides can't be used as a source of electrophile in Friedel-craft reaction
B) Polyacylation not possible
C) Nitrobenzene can undergo halogenation
D) Rearrangement product is the major product

38. Fraction left after 20 months $= \frac{400}{500} = 0.8.$

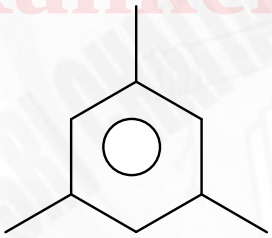
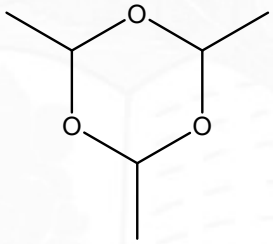
$\because\ (1-0.488) = 0.512 = (0.8)^3 \Rightarrow$ time required

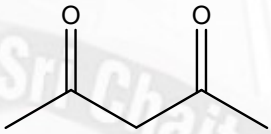
To reach this concentration is $3 \times 20 = 60$ month.

39. $\Rightarrow K_4\left[Fe(CN)_6\right] + KNO_2 + CH_3COOH \rightarrow K_3\left[Fe(CN)_6\right] + CH_3COOK + NO(g)$

$\Rightarrow NO + XeF_2 \rightarrow NOF + Xe$

$\Rightarrow 2\left[Fe(CN)_6\right]^{3-} + 3I^-(aq) \xrightarrow{H^+(aq)} 2\left[Fe(CN)_6\right]^{4-} + I_3^-(aq)$

40. P is  , Q is  , R is $Ph-CH=CH-\overset{\overset{O}{\|}}{C}-Ph$

S is 

41. Na_2CO_3 does not decompose on heating but it melts.

42. $\Rightarrow 2Na\left[BH_4\right] + I_2 \xrightarrow{\text{Diglyme}} 2NaI + B_2H_6 + H_2$

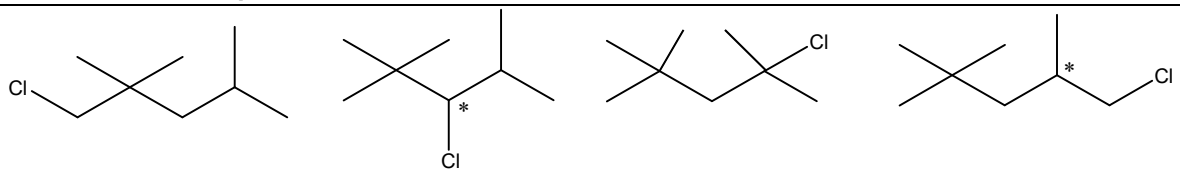
$\Rightarrow B_2H_6 + 6CH_3OH \longrightarrow 2B(OMe)_3 + 6H_2$

44. $X = \frac{1}{3} mol \quad CHI_3 = \frac{394}{3} = 131.33\,g$

$Y = 1\,mol \quad CH_3-\overset{\overset{O}{\|}}{C}-CH_2Br = 137\,g$

45. X is isooctane

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(1) (2) (1) (2)

$Y = 16, Z = 4$

46. $BrO_3^- + AsO_2^- + H_2O \rightarrow Br^- + AsO_4^{3-} + H^+$

m equation of excess $BrO_3^- = 2 \times 5 \times \frac{1}{25} = \frac{2}{5} m eqs$

n moles of excess $BrO_3^- = \frac{2}{5} \times \frac{1}{6} = \frac{1}{15} m mol$

m moles of BrO_3^- used in 1st reaction $= 20 \times \frac{1}{60} - \frac{1}{15} = \frac{4}{15} m mol$

$SeO_2^{3-} + BrO_3^- + H^+ \rightarrow SeO_4^{2-} + Br_2 + H_2O$

$m eqs\left(SeO_2^{3-}\right) + m eqs\left(BrO_3^-\right) = \frac{4}{15} \times 5 = \frac{4}{3} m eqs$

$m\ mol\ SeO_2^{3-} = \frac{4}{3} \times \frac{1}{2} = \frac{2}{3} m\ mol$

$\therefore m_{SeO_3^{2-}} = \frac{2}{3} \times 126 = 84 mg$

47. AB is isobaric process

$\therefore \frac{T_B}{T_A} = \frac{V_B}{V_A} = 2$

$\Rightarrow T_B = 2T_A = 600 K$

$Q_{AB} = \Delta H_{AB} = {}^n C_{p_m} \Delta T = 2 \times \frac{5R}{2} \times (600 - 300) = 3000 Cal$

BC is isothermal process

$Q_{BC} = nRT_B \ln\left(\frac{V_C}{V_B}\right)$

$= 2 \times 2 \times 600 \ln 2$

$= 2400 \ln 2 cal$

$Q_{CD} = {}^n C_{V_m} \Delta T = 2 \times \frac{3R}{2} \times (300 - 600) = -1800 cal$

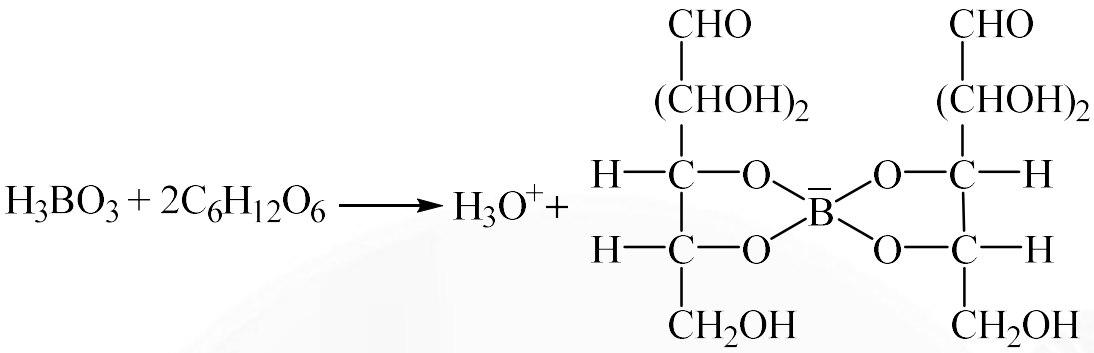
$Q_{DA} = nRT_D \ln\left(\frac{V_A}{V_D}\right) = 2 \times 2 \times 300 \ln\left(\frac{1}{4}\right) = -2400 \ln 2 cal$

$\therefore Q_{cycle} = 1200 cal.$

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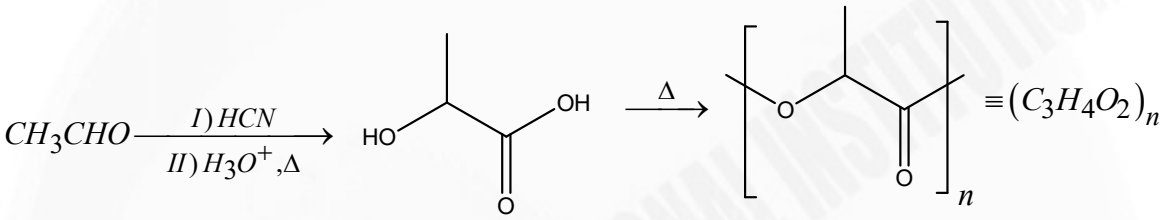
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48, 49.



$\Rightarrow x=2, y=4$

43.



$C_3H_6O_3$ Polylaclic acid

50. Initial mass of D_2SO_4 is $2.5\times1.40\times10^3\times0.5=1750\,g=M_{H_2O}$

The net cell reaction (involving $2e^-$ transfer) is :

$Pb(s)+PbO_2(s)+2D_2SO_4(aq)\rightarrow 2PbSO_4(s)+2D_2O(l)$

Before 1750

After $(1750-100n)$ $20n$

Mass of D_2SO_4 after discharge = $1750-100n$

Mass of solution after discharge = $(1750-100n+20n+1750)$

$$\%D_2SO_4=\frac{1750-100n}{3560-80n}=0.2$$

$\Rightarrow n=12.5.$

$n_{sol}=2500\,g$

$V_{sol}=2\,L$

$$n_{H_2SO_4}=\frac{1750-100\times12.5}{100}=5\,moles$$

$$\therefore [D_2SO_4]=\frac{5}{2}M.$$

51. 1 F is consumed per mol D_2SO_4.

∴ Charge delivered = 12.5 F.

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